Periodic Geometry and Fixed Counts for Cyclic Rule 184

Route-A structural certificate C175

Abstract

For every ring length and every conserved particle number, we classify the periodic states of cyclic Rule 184. Below half density they are exactly the binary words without adjacent particles; above half density they are exactly the words without adjacent holes. The dynamics is a rotation on this core, which yields a closed fixed-count formula for every iterate.

Keywords: Rule 184; traffic cellular automaton; periodic point; cyclic independent set; Artin–Mazur zeta.

1 Periodic-core theorem

On

$$X_{N,k} = \{x \in \{0,1\}^{\mathbb{Z}/N\mathbb{Z}} : \sum_i x_i = k\}$$

let

$$F(x)_i = x_{i-1}(1 - x_i) + x_i x_{i+1}.$$

This is the simultaneous traffic move $10 \mapsto 01$. Put $m = \min(k, N-k)$ and let $(\rho x)_i = x_{i-1}$. If $k \le N/2$, the periodic points are exactly the words with no cyclic 11, and $F = \rho$ there. If $k \ge N/2$, the periodic points are exactly the words with no cyclic 00, and $F = \rho^{-1}$ there. Thus every least period divides N.

At half filling the two descriptions agree and leave exactly the two alternating states, both of period two. For $k = 0$ and $k = N$, the unique uniform state is fixed.

2 Finite attraction by gap defects

We prove that the displayed core is complete. In the low-density branch, label the $m = k$ particles in cyclic order and let g_i be the number of holes between particle i and its successor. With $u_i = \mathbf{1}_{g_i>0}$, simultaneous motion gives

$$g'_i = g_i - u_i + u_{i+1}.$$

A new zero at index i occurs exactly when the old $g_{i+1} = 0$ and $g_i \le 1$. Hence every old zero-gap marker shifts one index backward unless its predecessor is at least two, in which case the marker is absorbed. The defect count $Z(g) = \#\{i : g_i = 0\}$ never increases.

If $Z(g) > 0$, then $\sum_i g_i = N - k \ge k = m$, so at least one gap is at least two. An excess gap remains at its index until a zero arrives immediately after it, whereas every unabsorbed zero shifts backward once per update. Within at most m steps one zero is absorbed. Repeating at most m times shows that the no-11 core is reached within m^2 updates. There every particle moves and $F = \rho$.

Above half density, holes move left. Exchanging particles and holes and reversing orientation gives the same gap proof and the no-00 core. If a periodic state entered this forward-invariant core after positive time, a sufficiently large multiple of its period would place the original state in the core. Thus no transient state can be periodic, completing the classification.

3 Every-iterate fixed count

Let

$$I(g,r) = \begin{cases} 1, & r = 0, \\ \dfrac{g}{g-r}\binom{g-r}{r}, & 1 \le r \le \lfloor g/2 \rfloor, \\ 0, & \text{otherwise.} \end{cases}$$

This is the number of cyclic length-g binary words with r isolated marked sites. For $n \ge 1$, set $g = \gcd(N, n)$ and $q = N/g$. Then

$$\#\operatorname{Fix}(F^n | X_{N,k}) = \begin{cases} I(g, m/q), & q \mid m, \\ 0, & q \nmid m. \end{cases}$$

An F^n-fixed word lies in the periodic core. There F^n is rotation by n in one of the two directions. Rotation invariance makes the word q repetitions of a cyclic block of length g; the block contains m/q isolated minority symbols, proving the formula.

4 Route-A boundary

The periodic cycles are intrinsic but have no proved arithmetic semantics. The exact source formula is not a target divisor comparison. Accordingly the preliminary v0.2 tuple is

$$(\texttt{A0\_FAIL}, \texttt{A1\_WEAK}, \texttt{A2\_FAIL}, \texttt{A3\_FAIL}, \texttt{A4\_FORMAL\_HINT}),$$

with overall `ROUTE_A_REJECTED` and Route B false. No target table, arithmetic local factor, Euler factor, root number, automorphy input, or Hilbert–Pólya claim is used. Scope: `NO_BAD_EULER_OR_ROOT_NUMBER`. The attraction argument also shows a sharp operator boundary. Every word in a sector has isolated minority symbols exactly when $m \leq 1$, so precisely those whole sectors are rotation permutations. For $m \geq 2$, a transient state exists; the finite whole-sector map is nonbijective and its uniform Koopman composition operator is not unitary. Restriction to the canonical periodic core is unitary but discards the transient dynamics.